| Reg. No. | | | | | | | | | |
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# MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

*(A constituent unit of MAHE, Manipal)*

## SEVENTH SEMESTER

## END SEMESTER EXAMINATIONS, NOVEMBER/DECEMBER 2024

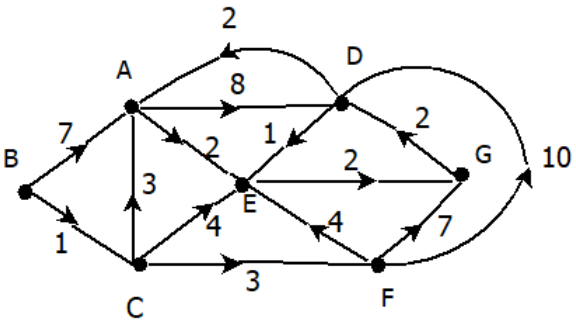
## Graphs & Matrices [MAT 4054]

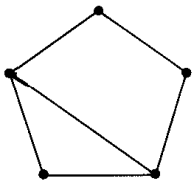
REVISED CREDIT SYSTEM

**Time: 3 Hours** **Date: 27 November 2024** **Max. Marks: 50**

**Instructions to Candidates:**

- Answer **ALL** the questions.

| Q.NO | Questions | Marks | CO | BTL |
|---|---|---|---|---|
| 1A. | Show that every self-complementary graph has either 4n or 4n+ 1 points. | 3 | 1 | 4 |
| 1B. | Let G be a connected graph. If diam(G) ≥3, then show that $diam(\bar{G}) \leq 3$. | 3 | 1 | 4 |
| 1C. | Use Dijkstra's algorithm to find shortest paths from B to all other vertices of the graph   | 4 | 2 | 3 |
| 2A. | Let G be a connected graph and v be a cut vertex of G. Then show that v is not a cut vertex of complement of G | 3 | 1 | 4 |
| 2B. | Let G be a connected cubic graph. Show that G has a cut vertex if and only if it has a bridge. | 3 | 1 | 4 |
| 2C. | Let G be a graph with n vertices, m edges and let $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ be the eigen values of G. Then show that $\lambda_1 \leq \sqrt{\frac{2m(n-1)}{n}}$ | 4 | 4 | 4 |
| 3A. | Show that a simple (p, q) graph G is a tree if and only it is connected and p = q + 1 | 3 | 2 | 4 |
| 3B. | Draw a graph For which $k(G) = 2,\ \lambda(G) = 3, \delta(G) = 4,$ $where\ k(G), \lambda(G)\ and\ \delta(G)$ respectively denote vertex connectivity, edge connectivity and minimum degree of the graph G. | 3 | 3 | 4 |

| | | | | |
|---|---|---|---|---|
| **3C.** | Let G be a graph with {0,1,-1} incidence matrix Q. Show that Q is totally unimodular. | ***4*** | ***5*** | ***4*** |
| **4A.** | Show that Laplacian matrix of a graph G is a symmetric positive semidefinite matrix. | ***3*** | ***5*** | ***3*** |
| **4B.** | Show that the graphs $K_5$ and $K_{3,3}$ are not planar graphs | ***3*** | ***3*** | ***4*** |
| **4C.** | Obtain chromatic polynomial of the following graph  Use the recurrence relation f(G, t) = f(G + e, t) + f(G.e, t) | ***4*** | ***3*** | ***4*** |
| **5A.** | Obtain eigen values of $K_n$. Find its energy. Hence conclude that any even positive integer is the energy of a graph. | ***3*** | ***4*** | ***4*** |
| **5B.** | For any graph G, show that $\chi(G) \leq 1 + \lambda_1(G)$, where $\chi(G)$ denotes the chromatic number and $\lambda_1(G)$ denotes the largest eigen value of the adjacency matrix of G | ***3*** | ***4*** | ***4*** |
| **5C.** | Let G be a connected graph with n vertices and let M be the {0,1} incidence matrix of G. Show that rank of M is n-1, if G is bipartite and n, otherwise. | ***4*** | ***5*** | ***3*** |